

Topic to discuss

- Milne's Predictor Corrector Method
- Numerical Problem
- Homework Problem

Milne's Predictor Corrector Method :

This method also use to solve ordinary differential Equation (ODE)

It belongs to family of multi-step methods:
Predictor corrector Methods.

Given the differential Equation,

$$\frac{dy}{dx} = f(x, y) \text{ and}$$

initial values known y_0, y_1, y_2, y_3

So we need to find function values

$$f_0, f_1, f_2, f_3$$

According to Milne's predictor,

$$y_{n+1}^{(p)} = y_{n-3} + \frac{4h}{3} (2f_{n-2} - f_{n-1} + 2f_n)$$

for $n=3$

$$y_4^{(p)} = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$$

Then calculate, f_4 means

$$f_{n+1} = f(x_{n+1}, y_{n+1}^{(p)})$$

According to Milne's Corrector,

$$y_{n+1}^{(c)} = y_{n-1} + \frac{h}{3} (f_{n-1} + 4f_n + f_{n+1})$$

for $n=3$

$$y_4^{(c)} = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

Repeat the correct

- Substitute corrected value again
- Recalculate f_{n+1}
- Apply corrector until the value remains unchanged.

* It is also called as Milne-Simpson or
Milne-Simpson Predictor Corrector Method.

Q: Compute $y(0.4)$ by Milne's Predictor-Corrector Method from the equation $\frac{dy}{dx} = xy + y^2$

Given that $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2773$

Solution: Given,

$$\frac{dy}{dx} = xy + y^2$$

$$h = 0.1$$

$$x_0 = 0, \quad y_0 = 1$$

$$x_1 = 0.1, \quad y_1 = 1.1169$$

$$x_2 = 0.2, \quad y_2 = 1.2773$$

We need to find y_3 first using RK4 method.

Step 1: finding y_3 using RK4

$$x_2 = 0.2 \quad y_2 = 1.2773$$

$$f(x, y) = xy + y^2$$

Calculating k_1, k_2, k_3, k_4 & y_3

$$\begin{aligned} k_1 &= h f(x_2, y_2) = 0.1 [0.2 \times 1.2773 + 1.2773^2] \\ &= 0.188695529 \end{aligned}$$

$$\begin{aligned} k_2 &= h f\left(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2}\right) \\ &= 0.1 f\left(0.2 + \frac{0.1}{2}, 1.2773 + \frac{0.188695529}{2}\right) \\ &= 0.22243295 \end{aligned}$$

$$K_3 = h f \left(x_2 + \frac{h}{2}, y_2 + \frac{K_2}{2} \right)$$
$$= 0.227510712$$

$$K_4 = h f(x_2 + h, y_2 + K_3)$$
$$= 0.271589849$$

Now,

$$y_3 = y_2 + \frac{1}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$
$$= 1.503995452$$

Now, we have 4 starting values:

$$y_0, y_1, y_2, y_3$$

finding value of f_0, f_1, f_2, f_3

We have, $f(x, y) = xy + y^2$

$$x_0 = 0, y_0 = 1$$

$$\Rightarrow f_0 = 0 \times 1 + 1^2 = 1$$

$$x_1 = 0.1, y_1 = 1.1169$$

$$\Rightarrow f_1 = 0.1 \times 1.1169 + 1.1169^2 = 1.35915561$$

$$x_2 = 0.2, y_2 = 1.2773$$

$$\Rightarrow f_2 = 0.2 \times 1.2773 + 1.2773^2 = 1.8869552$$

$$x_3 = 0.3, y_3 = 1.503995452$$

$$\Rightarrow f_3 = 0.3 \times 1.503995 + 1.503995^2 = 2.71320095$$

Now Applying Milne's Predictor Corrector Method to find y_4

Predictor:

$$y_4^{(p)} = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$$

$$= 1 + \frac{4 \times 0.1}{3} [2 \times 1.35915561 - 1.8869552 + 2 \times 2.71320095]$$

$$= 1.834367712$$

$$\begin{aligned}
 \text{Now, } f_4 &= f(0.4, 1.834367712) \\
 &= 0.4 \times 1.834367712 + (1.834367712)^2 \\
 &= 4.098651986
 \end{aligned}$$

Сопместор :

$$\begin{aligned}
 y_4^{(c1)} &= y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4) \\
 &= 1.2773 + \frac{0.1}{3} [1.8869552 + 4 \times 2.71320095 + 4.098651986] \\
 &= 1.838580370
 \end{aligned}$$

Again calculating f_4

$$f_4 = f(x_4, y_4^{(c1)})$$

$$= f(0.4, 1.838580370)$$

$$= 0.4 \times 1.838580370 + 1.838580370^2$$

$$= 4.115810570$$

Again correcting $y^{(c2)}$,

$$y_4^{(c2)} = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

$$= 1.839152301$$

$$\text{So, } y(0.4) = 1.8392 \text{ Ans.}$$

Homework Problem:

Q: Apply Milne's method to find $y(0.8)$ for the equation $\frac{dy}{dx} = x + y^2$, given that

$$y(0) = 0, \quad y(0.2) = 0.02, \quad y(0.4) = 0.0805, \quad y(0.6) = 0.1839$$

Solution: We have,

$$\begin{aligned} x_0 = 0, \quad y_0 = 0 &\rightarrow f_0 = 0 + 0^2 = 0 \\ x_1 = 0.2, \quad y_1 = 0.02 &\rightarrow f_1 = 0.2 + 0.02^2 = 0.2004 \\ x_2 = 0.4, \quad y_2 = 0.0805 &\rightarrow f_2 = 0.4 + 0.0805^2 = 0.40648025 \\ x_3 = 0.6, \quad y_3 = 0.1839 &\rightarrow f_3 = 0.6 + 0.1839^2 = 0.63381921 \end{aligned}$$

Now using Milne's Predictor Corrector formula,
Predictor:

$$\begin{aligned} y_4^{(P)} &= y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3) \\ &= 0 + \frac{4 \times 0.2}{3} [2 \times 0.2004 - 0.40648025 + 2(0.63381921)] \\ &= 0.336005445 \end{aligned}$$

Now, calculating, f_4

$$\begin{aligned} f_4 &= f(x_4, y_4) = f(0.8, 0.336005445) \\ &= 0.8 + 0.336005445^2 \\ &= 0.91289965 \end{aligned}$$

Correction:

$$\begin{aligned} y_4^{(c1)} &= y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4) \\ &= 0.0805 + \frac{0.2}{3} [0.40648025 + 4(0.63381921) + 0.91289965] \\ &= 0.309610047 \end{aligned}$$

Again correction:

$$\begin{aligned} f_4 &= f(0.8, 0.309610047) \\ &= 0.8 + 0.309610047^2 \\ &= 0.895858389 \end{aligned}$$

Again,

$$y_4^{(c2)} = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4)$$

$$= 0.0805 + \frac{0.2}{3} [0.40648025 + 4(0.63381921) + 0.895858389]$$

$$= 0.308473297$$

$$\text{So, } y(0.8) = 0.3085$$

Ans.

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